

## N. 1 (a)

$$\lim_{n \rightarrow \infty} \sqrt[n]{\frac{2^n + 3^n}{n^2 + n^3}} \Rightarrow \lim_{n \rightarrow \infty} \left( \frac{2^n + 3^n}{n^2 + n^3} \right)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left( \frac{3^n \left( \frac{2}{3} + 1 \right)}{n^3 \left( \frac{1}{n} + 1 \right)} \right)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} 3 \left( \frac{\frac{2}{3} + 1}{n^3 \left( \frac{1}{n} + 1 \right)} \right)^{\frac{1}{n}} \rightarrow 3$$

$$(b) \lim_{n \rightarrow \infty} \sqrt[n]{(n+1)^n - n^n} = \lim_{n \rightarrow \infty} \sqrt[n]{n^n \left[ \left(1 + \frac{1}{n}\right)^n - 1 \right]} = \lim_{n \rightarrow \infty} n \sqrt[n]{\left(1 + \frac{1}{n}\right)^n - 1} \geq n \sqrt[n]{1} = +\infty$$

$$(c) \lim_{n \rightarrow \infty} \frac{\sqrt{9n+1} - \sqrt{9n+8}}{\sqrt{4n-1} - \sqrt{4n+7}} = \frac{\sqrt{9n+1} - \sqrt{9n+8}}{\sqrt{4n-1} - \sqrt{4n+7}} \cdot \frac{\sqrt{9n+1} + \sqrt{9n+8}}{\sqrt{9n+1} + \sqrt{9n+8}} \cdot \frac{\sqrt{4n-1} + \sqrt{4n+7}}{\sqrt{4n-1} + \sqrt{4n+7}} = \frac{(9n+1) - (9n+8)}{(4n-1) - (4n+7)} \cdot \frac{\sqrt{9n+1} + \sqrt{9n+8}}{\sqrt{4n-1} + \sqrt{4n+7}} =$$

$$= \frac{7}{8} \left( \frac{2\sqrt{n} \left( \sqrt{1 - \frac{1}{4n}} - \sqrt{1 - \frac{3}{4n}} \right)}{3\sqrt{n} \left( \sqrt{1 + \frac{1}{4n}} - \sqrt{1 + \frac{3}{4n}} \right)} \right) = \frac{7}{8} \cdot \frac{2}{3} = \frac{14}{24} = \frac{7}{12}$$

$$(d) \lim_{n \rightarrow \infty} \frac{(2n+3)! - (2n)!}{n^2((2n+1)! - (2n)!)} = \frac{(2n+3)(2n+2)(2n+1)(2n)! - (2n)!}{n^2((2n+1)(2n)! - (2n)!)} = \frac{(2n+3)(2n+2)(2n+1) - 1}{n^2((2n+1) - 1)} = \frac{n^3 \left( 2 + \frac{3}{n} \right) \left( 2 + \frac{2}{n} \right) \left( 2 + \frac{1}{n} \right) - 1}{2n^3} = \frac{8}{2} = 4$$

$$(e) \lim_{n \rightarrow \infty} \frac{n^n}{4^n \cdot n!} \stackrel{C.R.}{=} \frac{(n+1)^{n+1}}{4^{n+1} \cdot (n+1)!} \cdot \frac{4^n \cdot n!}{n^n} = \frac{(n+1)^n (n+1)}{4^n \cdot 4 \cdot (n+1) n!} \cdot \frac{4^n \cdot n!}{n^n} = \frac{1}{4} \left( 1 + \frac{1}{n} \right)^n = \frac{e}{4}$$

$$(f) \lim_{n \rightarrow \infty} \frac{(2n)!}{n! \cdot n^n} = \frac{2(n+1)!}{(n+1)! (n+1)^{n+1}} \cdot \frac{n! \cdot n^n}{(2n)!} = \frac{(2n+2)(2n+1)(2n)!}{(n+1)(n+1)(n+1)^n} \cdot \frac{n! \cdot n^n}{(2n)!} = \frac{n^2 \left( 2 + \frac{2}{n} \right) \left( 2 + \frac{1}{n} \right)}{(n+1)^2 (n+1)^n} \cdot n^n \rightarrow \frac{4}{e} > 1$$

$$(n) \lim_{n \rightarrow \infty} n \left( \sqrt[n]{n+1} - 1 \right) = n \left( \sqrt[n]{\frac{n+1}{n+6}} - 1 \right) = \frac{\left( 1 + \frac{-5}{n+6} \right)^{\frac{1}{n}} - 1}{\frac{-5}{n+6}} \cdot \frac{-5n}{n+6} \rightarrow -\frac{5}{4}$$

Ricorda:

$$\lim_{x \rightarrow 0} \frac{(1+x)^a - 1}{x} = a$$

$$(g) \lim_{n \rightarrow \infty} n^2 \ln(\cos(\frac{1}{n})) = \frac{\ln(1 + \cos(\frac{1}{n}) - 1)}{\cos(\frac{1}{n}) - 1} \cdot \frac{\cos(\frac{1}{n}) - 1}{\frac{1}{n^2}} \rightarrow -\frac{1}{2}$$

## N. 2

$$(a) \lim_{n \rightarrow \infty} \left( 1 - \frac{2}{\sqrt{n}} \right)^n = e^{\frac{n \log \left( 1 - \frac{2}{\sqrt{n}} \right)}{-\frac{2}{\sqrt{n}}}} \cdot \left( -\frac{2}{\sqrt{n}} \right) \rightarrow 0$$

$$(b) \lim_{n \rightarrow \infty} \left( 1 + \frac{2}{n} \right)^{\sqrt{n}} = \exp \left( \underbrace{\sqrt{n}}_{\rightarrow 0} \cdot \underbrace{\ln \left( 1 + \frac{2}{n} \right)}_{\rightarrow 0} \cdot \underbrace{\left( \frac{1}{\sqrt{n}} \right)}_{\rightarrow 0} \right) \rightarrow e^0 = 1$$

$$(c) \lim_{n \rightarrow \infty} \left( 1 + \frac{1}{2^{n-1}} \right)^{2^{n+1}} = \lim_{n \rightarrow \infty} \left( \left( 1 + \frac{1}{2^{n-1}} \right)^{2^{n-1}} \right)^4 \rightarrow e^4$$

$$(d) \lim_{n \rightarrow \infty} \left( 1 + \frac{3}{\log(n)} \right)^{\log(\sqrt{n})} = \text{Per } n \rightarrow \infty$$

$$\left( 1 + \frac{3}{2 \log(n)} \right)^{\frac{1}{2} \log(n)} = \left( \left( 1 + \frac{3}{2 \log(n)} \right)^{\log(n)} \right)^{\frac{1}{2}} \rightarrow (e^{\frac{3}{2}})^{\frac{1}{2}} = e^{\frac{3}{4}}$$

$$(e) \lim_{n \rightarrow \infty} \left( \frac{2n^2 + 5n + 4}{2n^2 - 3n + 6} \right)^{3n+2} = \exp \left[ 3n+2 \left[ \ln \left( 1 + \frac{8n-2}{2n^2-3n+6} \right) - \frac{8n-2}{2n^2-3n+6} \right] \right] \rightarrow e^{12}$$

$$\textcircled{f} \lim_{n \rightarrow \infty} \left( \frac{3^n + 2^{n+1}}{3^n \cdot 2^{n+1}} \right)^{(3/2)^n} = \lim_{n \rightarrow \infty} \frac{\left(1 + \frac{2}{(3/2)^n}\right)^{(3/2)^n}}{1 + \left(\frac{1/2}{(3/2)^n}\right)^{(3/2)^n}} \rightarrow \frac{e^2}{e^{1/2}} = e^{3/2}$$

N. 3

$$\textcircled{d} \lim_{x \rightarrow \infty} \frac{(1+2x^2)^x - 1}{(1+x)^2 + (1-x)^2 \cdot 2} = \frac{(1+2x^2)^x - 1}{6x^2} = \frac{(1+2x^2)^x \cdot 1}{6x^2} \rightarrow \frac{1}{3} \cdot \frac{(1+2x^2)^x - 1}{2x^2} \rightarrow \frac{\pi}{3}$$

$$(1+x)(1+2x+x^2) = 1 + 2x + x^2 + x + 2x^2 + x^3$$

$$1 + 3x + 3x^2 + x^3$$

$$\left. \begin{array}{l} 1 + 3x + 3x^2 + x^3 \\ (1-x)^3 = 1 + 3x^2 - 3x - x^3 \end{array} \right\} \rightarrow \cancel{1+3x+3x^2+x^3} + \cancel{1+3x^2-3x-x^3} = \cancel{2} - \cancel{2}$$

$$\textcircled{b} \lim_{x \rightarrow 0} \frac{9^x - 4^x}{\ln(2x)} = \left( \frac{9^x - 1}{x} - \frac{4^x - 1}{x} \right) \cdot \frac{x}{\ln(2x)} \rightarrow \ln(9) - \ln(4)$$

$$\text{Ricorda } \frac{a^x - 1}{x} = \ln(a)$$

$$\textcircled{c} \lim_{x \rightarrow 0} \frac{\log(1+2x)}{\sin(3^x-1)} = \frac{\log(1+2x)}{2x} \cdot \frac{x}{3^x-1} \cdot \frac{3^x-1}{\sin(3^x-1)} \rightarrow \frac{2}{\log(3)}$$

$$\textcircled{d} \lim_{x \rightarrow 0^+} \frac{\sin(2^x-1)}{\log(1+2x^2)} = \underbrace{\frac{\sin(2^x-1)}{2^x-1}}_{\rightarrow 1} \cdot \underbrace{\frac{2^x-1}{x}}_{\rightarrow \log(2)} \cdot \underbrace{\frac{1}{2x}}_{\rightarrow +\infty} \cdot \underbrace{\frac{2x^2}{\log(1+2x^2)}}_{\rightarrow 1} \rightarrow +\infty$$

N. 5

$$\textcircled{a} \forall n \in \mathbb{N}^+$$

$$n! \leq n^n \leq 2^n \cdot (n!)^2$$

$$\text{P.B. } 1! \leq 1^1 \leq 2^1 \cdot (1!)^2 \Rightarrow 1 \leq 1 \leq 2 \text{ OK}$$

$$\textcircled{1} n! \leq n^n \Rightarrow n! = n \cdot (n-1) \cdot (n-2) \dots (1)$$

$$\text{allora } n! \leq n^n$$

$$n^n = n \cdot n \cdot n \cdot n \dots n \cdot n$$

$$\textcircled{2} n^n \leq 2^n \cdot (n!)^2 \Rightarrow (n+1)^{n+1} \leq 2^n \cdot (n+1)!^2 \Rightarrow (n+1)^n (n+1) \leq 2^n ((n+1)!)^2 (n+1)$$

$$(n+1) \cdot e \cdot 2^n (n!)^2 \leq 2^{n+1} ((n+1)!)^2 \Rightarrow e < 4 \leq 2(n+1)$$

$\textcircled{b}$  Calcolare i seguenti limiti:

$$\lim_{n \rightarrow \infty} (n!)^{1/n^2} \Rightarrow 1 \leq (n!)^{1/n^2} \leq (n^n)^{1/n^2} = n^{1/n} \rightarrow 1 \text{ D.C. lim } \rightarrow 1$$

$$\lim_{n \rightarrow \infty} (n!)^{1/n} \Rightarrow (n!)^{1/n} \geq \left(\frac{n^n}{2^n}\right)^{1/2n} = \sqrt{n/2} \rightarrow \infty$$